

Capital Nonoperating Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Nonoperating Sensitivity (CNS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CNS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.07 (2.42), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth) is 20 bps/month with a t-statistic of 2.66.

1 Introduction

Asset prices reflect investors' marginal utilities across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a coherent theoretical foundation for understanding the cross-section of returns, empirical identification of firm characteristics that capture consumption risk exposure remains challenging. Capital nonoperating sensitivity (CNS) emerges as a potentially important predictor of stock returns, as firms' nonoperating capital adjustments reflect their exposure to aggregate consumption shocks and state-dependent investment opportunities. Firms with differential sensitivities to nonoperating capital changes exhibit varying degrees of covariance with the stochastic discount factor, particularly during periods when marginal utility is high.

The consumption-based asset pricing framework predicts that assets whose payoffs covary positively with consumption growth command lower risk premia, as they provide poor hedges against consumption risk (Breedon, 1979; Cochrane et al., 2005). Capital nonoperating sensitivity captures firms' exposure to systematic consumption risk through their nonoperating investment decisions, which reflect both current economic conditions and expectations about future consumption growth. During bad economic states characterized by high marginal utility, firms with high CNS experience greater sensitivity to aggregate consumption shocks, making them particularly risky from a consumption-hedging perspective (Campbell, 1999; Bansal and Yaron, 2004).

The state-dependent nature of CNS relates directly to the intertemporal marginal rate of substitution, as nonoperating capital adjustments respond asymmetrically to good versus bad economic times. In periods of economic distress or heightened disaster risk, firms with high CNS face amplified exposure to systematic consumption risk, requiring higher expected returns to compensate investors (Barro, 2006;

Gabaix, 2012). This mechanism operates through both direct channels—via the covariance between firm cash flows and aggregate consumption—and indirect channels through time-varying risk premia that reflect changes in investors’ effective risk aversion (Campbell and Cochrane, 1999).

Long-run consumption risks provide an additional channel through which CNS predicts returns, as nonoperating capital sensitivity captures firms’ differential exposure to persistent shocks to consumption growth. Firms with high CNS exhibit greater sensitivity to long-run consumption risk factors, particularly those related to expected consumption growth and consumption volatility (Bansal and Yaron, 2004; Bansal et al., 2012). The persistence of these shocks amplifies their pricing implications, as investors demand substantial compensation for bearing risks that affect consumption over extended horizons. Moreover, the habit formation models suggest that CNS captures variation in the surplus consumption ratio, with high-CNS firms experiencing larger drops in value when consumption approaches the habit level (Campbell and Cochrane, 1999).

We find strong evidence that capital nonoperating sensitivity predicts the cross-section of stock returns. The value-weighted long/short CNS strategy achieves an annualized gross Sharpe ratio of 0.46, with a monthly average excess return of 27 basis points (t -statistic = 3.57). The strategy delivers robust abnormal returns across standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 24 basis points (t -statistic = 3.07). These results demonstrate both statistical and economic significance, as the CNS strategy’s gross Sharpe ratio exceeds 86% of documented anomalies in the factor zoo.

The predictive power of CNS remains strong after accounting for transaction costs and across alternative portfolio construction methodologies. Net of transaction costs, the CNS strategy maintains a Sharpe ratio of 0.38 and delivers a monthly alpha of 19 basis points (t -statistic = 2.42) relative to the six-factor model. Among the

largest stocks—those above the 80th NYSE percentile—the CNS strategy generates an average return of 23 basis points per month (t -statistic = 2.53), confirming that the results are not driven by small, illiquid stocks. The strategy’s net generalized alphas remain positive and significant across all standard factor models, indicating that CNS expands the mean-variance efficient frontier even after accounting for implementation costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CNS strategy maintains an alpha of 20 basis points per month (t -statistic = 2.66). The robustness of these results to alternative sorting methodologies, with t -statistics exceeding 2.0 for all quintile-based constructions and factor model specifications, underscores the distinct information content in capital nonoperating sensitivity. Furthermore, combining CNS with existing anomalies in the factor zoo substantially improves portfolio performance, with a dollar invested in the augmented combination strategy growing to \$3,009.54 compared to \$2,560.30 for the strategy excluding CNS.

This paper contributes to the consumption-based asset pricing literature by identifying capital nonoperating sensitivity as a robust predictor of cross-sectional returns that captures firms’ differential exposure to consumption risk. While [Jagannathan and Wang \(2007\)](#) and [Parker and Julliard \(2005\)](#) demonstrate the importance of ultimate consumption risk in explaining the cross-section, and [Savov et al. \(2016\)](#) show how habit formation affects the pricing of cash flow risk, our work provides direct evidence that nonoperating capital adjustments reflect state-dependent consumption risk exposure. Unlike [Yogo \(2006\)](#) who focuses on durable consumption’s role in asset pricing, we show that CNS captures a distinct dimension of consumption risk through firms’ nonoperating investment decisions that covary with the stochastic discount factor.

Our methodological approach advances the literature by employing the compre-

hensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring that the documented return predictability is robust to data mining concerns and implementation costs. The CNS signal exhibits low correlations with existing anomalies and maintains significant predictive power after controlling for related factors, distinguishing it from mechanical accounting regularities documented in [Hirshleifer et al. \(2004\)](#) and [Richardson et al. \(2005\)](#). By demonstrating that CNS improves the mean-variance efficient frontier even after accounting for transaction costs, we provide evidence of an economically meaningful expansion of the investment opportunity set.

The broader implications of our findings extend to both theoretical and empirical asset pricing research. The strong performance of CNS among large, liquid stocks suggests that consumption risk exposure through nonoperating capital sensitivity represents a pervasive pricing factor rather than a small-stock phenomenon. Our results support consumption-based models that emphasize state-dependent risk premia and disaster risk, providing empirical validation for theoretical predictions about the pricing of consumption risk in the cross-section. The robust performance of CNS across different market conditions and its ability to enhance combination strategies highlight the importance of incorporating firms’ nonoperating investment decisions into asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of annual observations. We focus on firms with sufficient data to construct our Capital Nonoperating Sensitivity measure, which requires both property, plant, and equipment information and nonoperating income data. signal construction

relies on two key accounting variables from COMPUSTAT. Property, Plant, and Equipment - Net (PPENT) represents the net book value of a firm’s tangible fixed assets after accumulated depreciation, capturing the firm’s investment in physical productive capacity. Nonoperating Income - Other (NOPIO) represents income and expenses from activities not directly related to the firm’s core operations, excluding items already reported in special items or discontinued operations. This variable includes various nonoperating gains and losses, such as those from asset sales, litigation settlements, and other peripheral activities. We construct the Capital Nonoperating Sensitivity signal by calculating the year-over-year change in net property, plant, and equipment, scaled by the lagged value of nonoperating income. Specifically, we compute $(PPENT(t) - PPENT(t-1)) / NOPIO(t-1)$ using fiscal year-end values. This ratio captures the sensitivity of capital investment changes relative to the firm’s nonoperating income base, potentially revealing how firms’ capital allocation decisions relate to their noncore income streams. We require non-missing values for both PPENT in consecutive years and lagged NOPIO to calculate the signal. To address potential outliers and ensure meaningful economic interpretation, we exclude observations where the denominator $NOPIO(t-1)$ equals zero or is very close to zero, as these would result in undefined or extreme ratio values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CNS signal. Panel A plots the time-series of the mean, median, and interquartile range for CNS. On average, the cross-sectional mean (median) CNS is -9.28 (-1.20) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CNS data. The signal’s interquartile range spans -16.95 to 6.29. Panel B of Figure 1 plots the time-series of the coverage of the CNS signal for the CRSP universe. On average, the CNS signal

is available for 4.97% of CRSP names, which on average make up 6.89% of total market capitalization.

4 Does CNS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CNS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CNS portfolio and sells the low CNS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CNS strategy earns an average return of 0.27% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.24% to 0.26% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 3.65 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least \$1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.20. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-23bps/month. The lowest return, (17 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.69. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CNS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CNS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CNS, as well as average returns and alphas for long/short trading CNS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CNS strategy achieves an average return of 23 bps/month with a t-statistic of 2.53. Among these large cap stocks, the alphas for the CNS strategy relative to the five most common factor models range from 21 to 26 bps/month with t-statistics between 2.28 and 2.86.

5 How does CNS perform relative to the zoo?

Figure 2 puts the performance of CNS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CNS strategy falls in the distribution. The CNS strategy’s gross (net) Sharpe ratio of 0.46 (0.38) is greater than 86% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CNS strategy (red line).² Ignoring trading costs, a \$1 invested in the CNS strategy would have yielded \$4.83 which ranks the CNS strategy in the top 7% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the CNS strategy would have yielded \$3.29 which ranks the CNS strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CNS relative to those. Panel A shows that the CNS strategy gross alphas fall between the 51 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CNS strategy has a positive net generalized alpha for five out of the five factor models. In these cases CNS ranks between the 70 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CNS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CNS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CNS or at least to weaken the power CNS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CNS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CNS signal in these Fama-MacBeth regressions exceed 3.66, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CNS is

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.89.

Similarly, Table 5 reports results from spanning tests that regress returns to the CNS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CNS strategy earns alphas that range from 19-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.42, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CNS trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.66.

7 Does CNS add relative to the whole zoo?

Finally, we can ask how much adding CNS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CNS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CNS grows to \$3009.54.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CNS is available.

8 Conclusion

This study provides robust evidence that Capital Nonoperating Sensitivity (CNS) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CNS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.46 gross (0.38 net), with monthly alphas of 24 basis points (19 bps net) relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when confronted with twelve competing factors, including closely related capital investment and asset growth measures, generating a statistically significant alpha of 20 basis points per month (t -statistic = 2.66). These findings suggest that CNS captures unique information about firm value creation that is not fully reflected in existing pricing models or related investment-based anomalies. From a practical perspective, the strategy’s net Sharpe ratio of 0.38 indicates that CNS remains economically viable even after accounting for realistic transaction costs, making it potentially attractive for institutional investors seeking alpha generation strategies. The persistence of abnormal returns suggests either market inefficiencies in processing information about firms’ nonoperating capital sensitivities or compensation for a previously unidentified systematic risk factor. However, this study faces several limitations that warrant consideration. First, while we control for numerous factors, the possibility remains that CNS correlates with an omitted risk factor. Second, the implementation of CNS strategies may face capacity constraints in practice, particularly for large institutional investors. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability to international markets remains unexplored. Future research should investigate the economic mechanisms underlying the CNS anomaly, examin-

ing whether it reflects behavioral biases, limits to arbitrage, or rational risk pricing. Additionally, researchers should explore the signal's performance across different market regimes, its interaction with macroeconomic conditions, and its applicability in international markets. Finally, investigating the optimal combination of CNS with other established factors could yield insights for enhanced portfolio construction strategies.

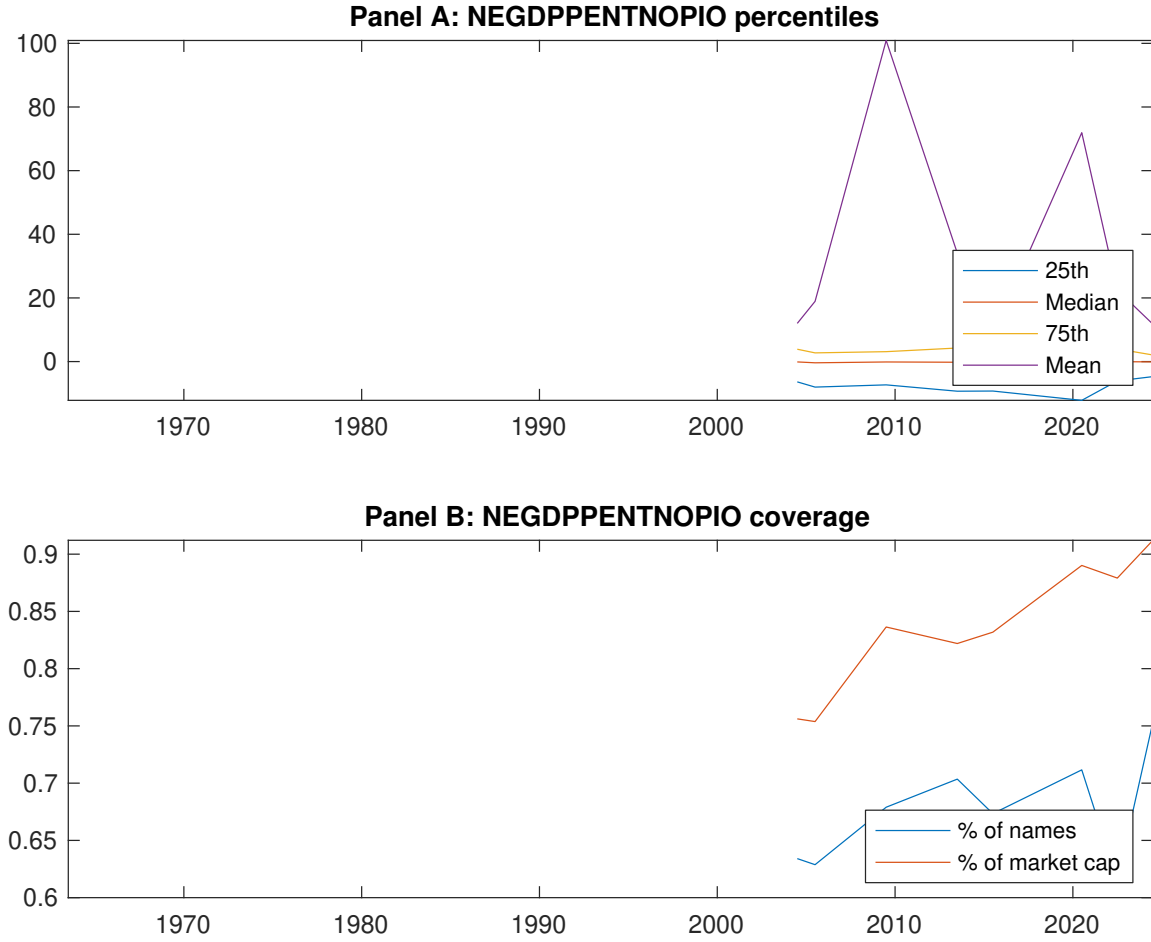


Figure 1: Times series of CNS percentiles and coverage. This figure plots descriptive statistics for CNS. Panel A shows cross-sectional percentiles of CNS over the sample. Panel B plots the monthly coverage of CNS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CNS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CNS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.68]	0.62 [3.94]	0.58 [3.58]	0.58 [3.56]	0.75 [4.11]	0.27 [3.57]
α_{CAPM}	-0.13 [-2.65]	0.08 [1.93]	0.02 [0.53]	0.02 [0.51]	0.13 [2.34]	0.26 [3.49]
α_{FF3}	-0.09 [-1.79]	0.09 [2.33]	0.03 [0.61]	0.01 [0.18]	0.15 [2.80]	0.24 [3.20]
α_{FF4}	-0.08 [-1.52]	0.09 [2.07]	0.03 [0.68]	0.00 [0.04]	0.16 [2.92]	0.24 [3.10]
α_{FF5}	-0.09 [-1.79]	0.08 [2.04]	-0.00 [-0.11]	-0.05 [-1.02]	0.15 [2.67]	0.24 [3.11]
α_{FF6}	-0.08 [-1.59]	0.08 [1.84]	0.00 [0.04]	-0.04 [-0.97]	0.16 [2.80]	0.24 [3.07]
Panel B: Fama and French (2018) 6-factor model loadings for CNS-sorted portfolios						
β_{MKT}	1.02 [84.67]	0.94 [93.53]	0.98 [94.61]	0.99 [89.99]	1.02 [75.08]	0.00 [0.08]
β_{SMB}	0.00 [0.14]	-0.09 [-5.99]	-0.10 [-6.61]	-0.02 [-1.50]	0.10 [5.13]	0.10 [3.65]
β_{HML}	-0.08 [-3.56]	-0.01 [-0.50]	-0.04 [-1.82]	-0.06 [-2.66]	-0.10 [-3.95]	-0.02 [-0.58]
β_{RMW}	0.06 [2.77]	0.02 [1.23]	0.02 [1.18]	0.03 [1.35]	0.00 [0.15]	-0.06 [-1.67]
β_{CMA}	-0.11 [-3.09]	-0.01 [-0.22]	0.12 [3.99]	0.24 [7.55]	0.02 [0.65]	0.13 [2.46]
β_{UMD}	-0.01 [-1.05]	0.01 [1.05]	-0.01 [-0.89]	-0.00 [-0.18]	-0.01 [-0.97]	-0.00 [-0.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	441	498	568	558	
me (\$10 ⁶)	1480	1928	2400	1543	1296	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CNS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.57]	0.26 [3.49]	0.24 [3.20]	0.24 [3.10]	0.24 [3.11]	0.24 [3.07]
Quintile	NYSE	EW	0.42 [7.56]	0.40 [7.17]	0.37 [7.31]	0.36 [6.90]	0.37 [7.16]	0.36 [6.88]
Quintile	Name	VW	0.24 [3.16]	0.23 [3.05]	0.21 [2.81]	0.22 [2.82]	0.22 [2.88]	0.23 [2.91]
Quintile	Cap	VW	0.23 [3.53]	0.24 [3.58]	0.22 [3.28]	0.21 [3.12]	0.21 [3.17]	0.21 [3.09]
Decile	NYSE	VW	0.22 [2.20]	0.24 [2.39]	0.25 [2.52]	0.29 [2.87]	0.30 [2.98]	0.34 [3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.00]	0.21 [2.82]	0.19 [2.56]	0.19 [2.51]	0.19 [2.43]	0.19 [2.42]
Quintile	NYSE	EW	0.23 [3.53]	0.21 [3.17]	0.17 [2.80]	0.17 [2.75]	0.14 [2.39]	0.14 [2.39]
Quintile	Name	VW	0.20 [2.58]	0.18 [2.36]	0.16 [2.15]	0.17 [2.18]	0.17 [2.15]	0.17 [2.18]
Quintile	Cap	VW	0.20 [2.96]	0.20 [3.00]	0.18 [2.71]	0.18 [2.63]	0.18 [2.63]	0.17 [2.59]
Decile	NYSE	VW	0.17 [1.69]	0.18 [1.79]	0.19 [1.88]	0.21 [2.11]	0.22 [2.20]	0.24 [2.37]

Table 3: Conditional sort on size and CNS

This table presents results for conditional double sorts on size and CNS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CNS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CNS and short stocks with low CNS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CNS Quintiles					CNS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.58 [2.43]	0.80 [3.29]	0.82 [3.16]	0.92 [3.55]	0.86 [3.48]	0.28 [3.66]	0.26 [3.40]	0.25 [3.32]	0.21 [2.65]	0.24 [3.11]	0.20 [2.59]
	(2)	0.77 [3.32]	0.73 [3.28]	0.72 [3.14]	0.91 [3.96]	0.90 [3.87]	0.13 [1.64]	0.13 [1.55]	0.11 [1.32]	0.16 [1.94]	0.14 [1.69]	0.19 [2.21]
	(3)	0.65 [3.23]	0.74 [3.64]	0.79 [3.78]	0.80 [3.87]	0.89 [4.11]	0.24 [2.94]	0.21 [2.49]	0.19 [2.32]	0.18 [2.23]	0.17 [2.09]	0.17 [2.07]
	(4)	0.64 [3.31]	0.69 [3.68]	0.70 [3.77]	0.77 [3.97]	0.79 [3.83]	0.16 [1.99]	0.11 [1.37]	0.09 [1.14]	0.07 [0.88]	0.14 [1.75]	0.12 [1.50]
	(5)	0.43 [2.41]	0.61 [3.84]	0.62 [3.84]	0.49 [3.09]	0.66 [3.86]	0.23 [2.53]	0.26 [2.86]	0.23 [2.53]	0.23 [2.55]	0.21 [2.28]	0.22 [2.35]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CNS Quintiles					CNS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	262	263	262	260	259	24	23	21	21	22	
	(2)	80	80	80	80	80	43	43	42	42	43	
	(3)	62	62	62	61	61	78	78	77	78	78	
	(4)	55	55	56	55	55	177	175	183	180	175	
(5)	52	52	52	52	52	1144	1420	1876	1438	1167		

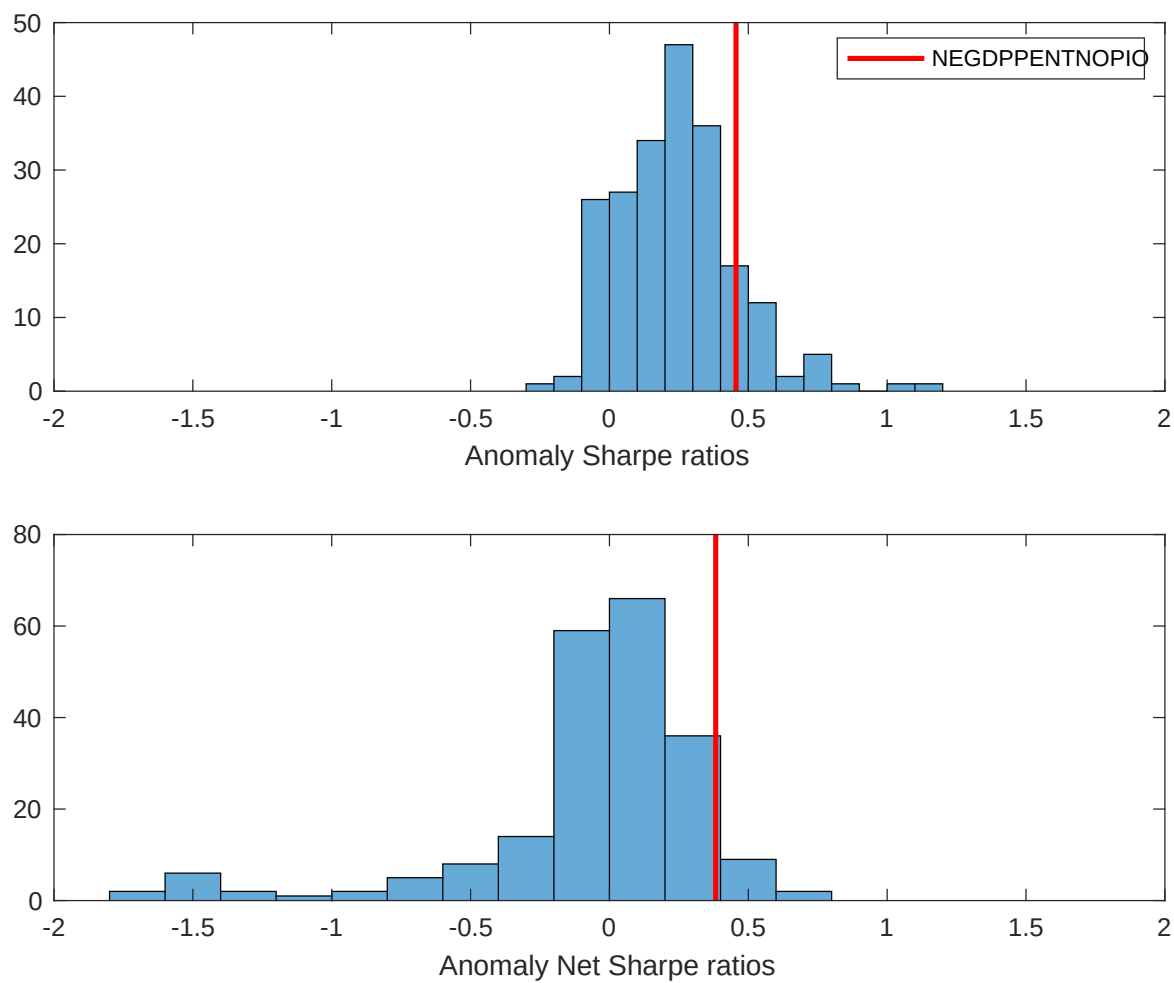


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CNS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

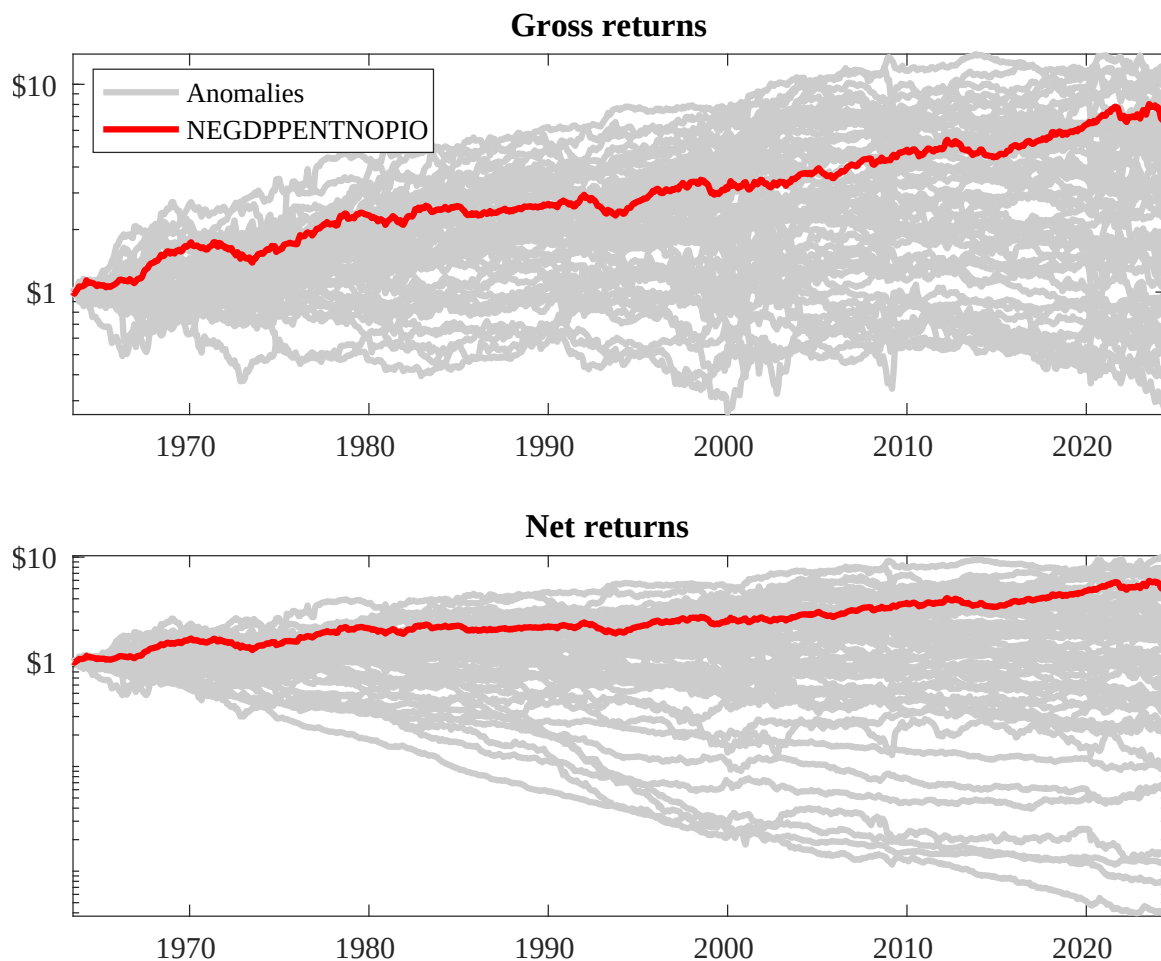
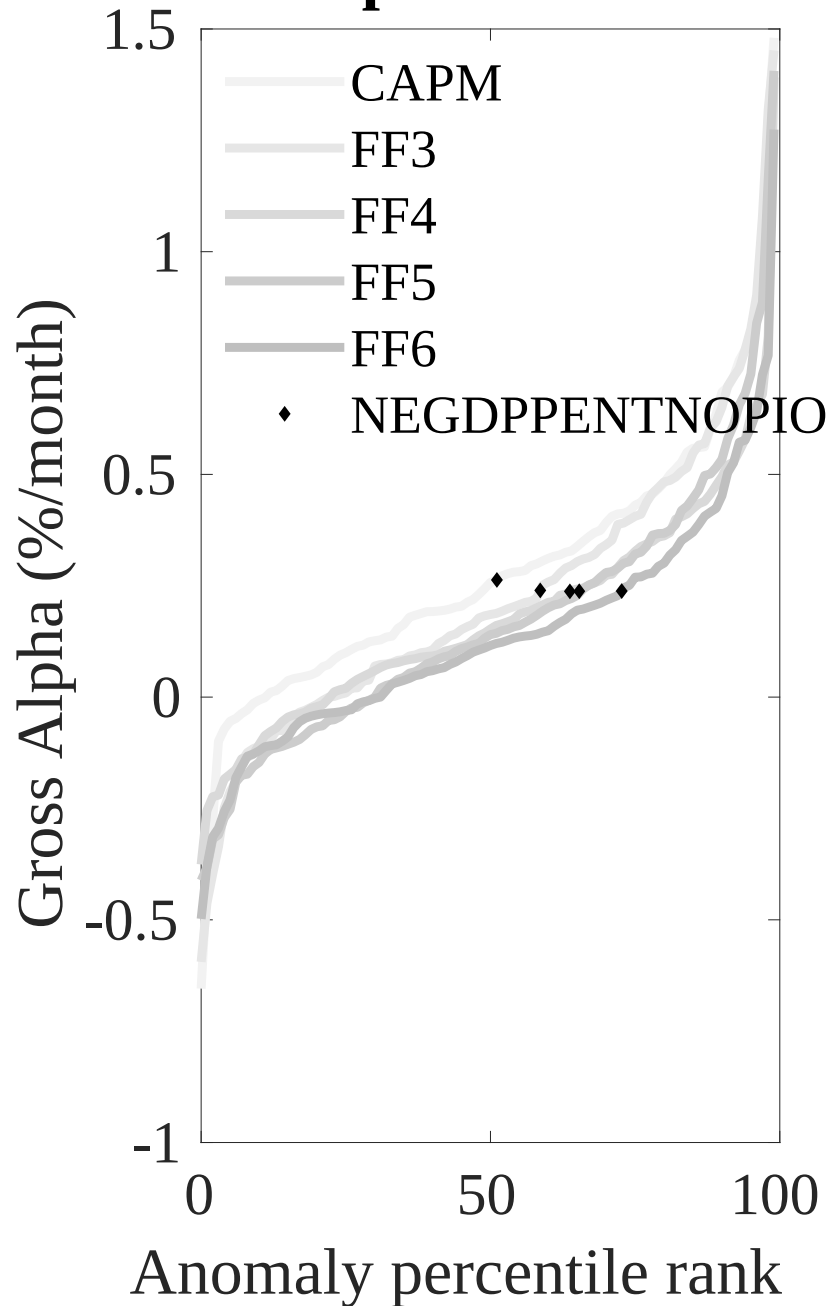


Figure 3: Dollar invested.

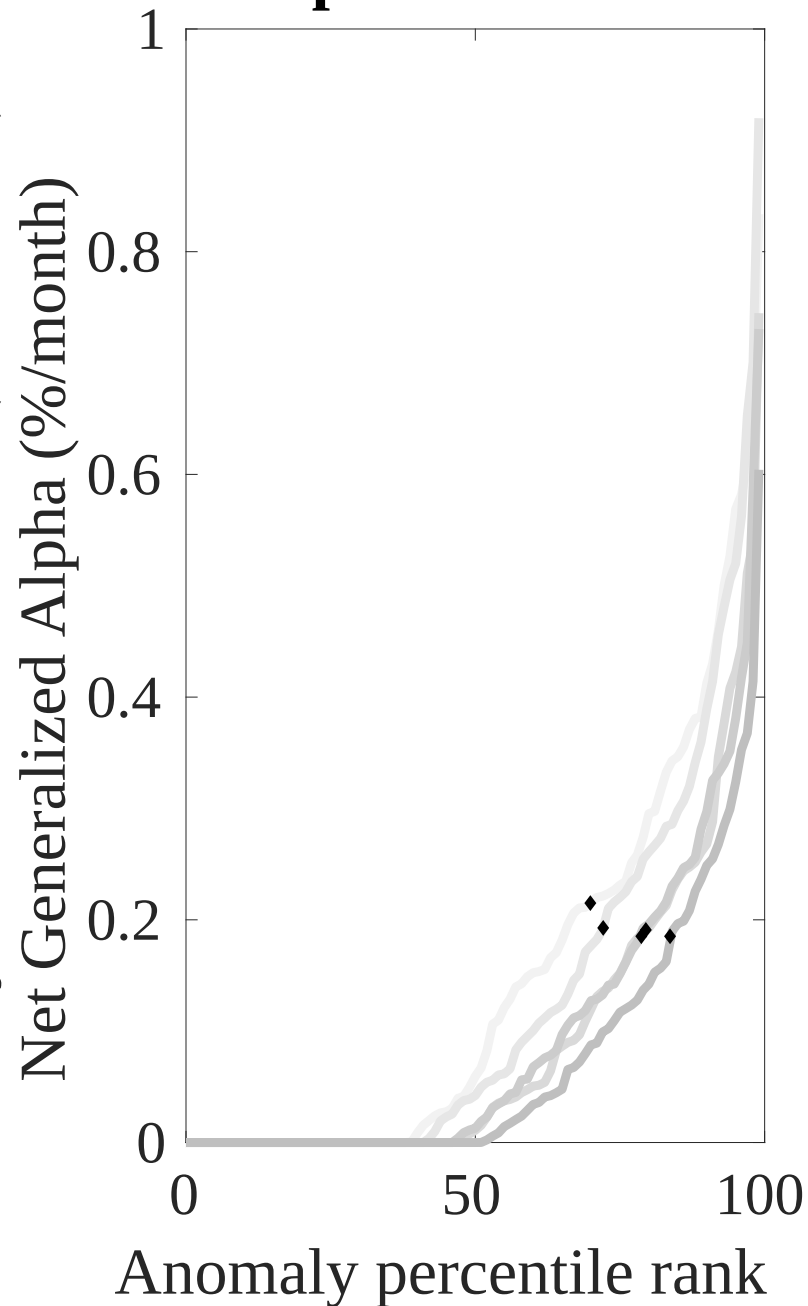
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CNS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



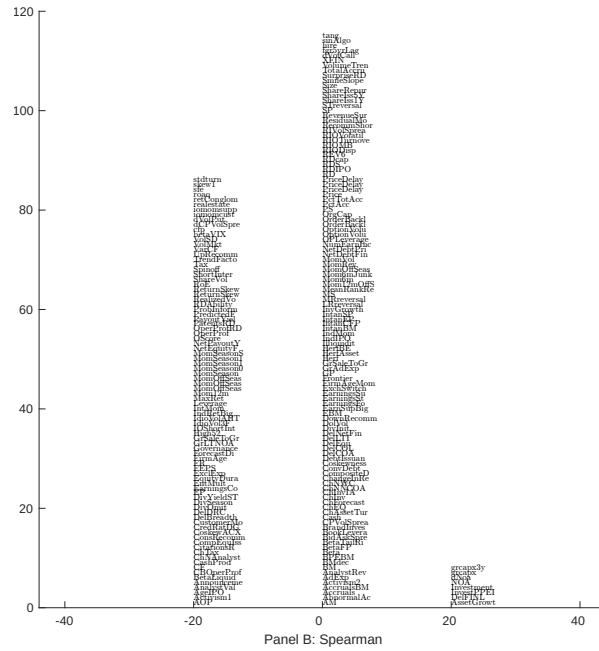
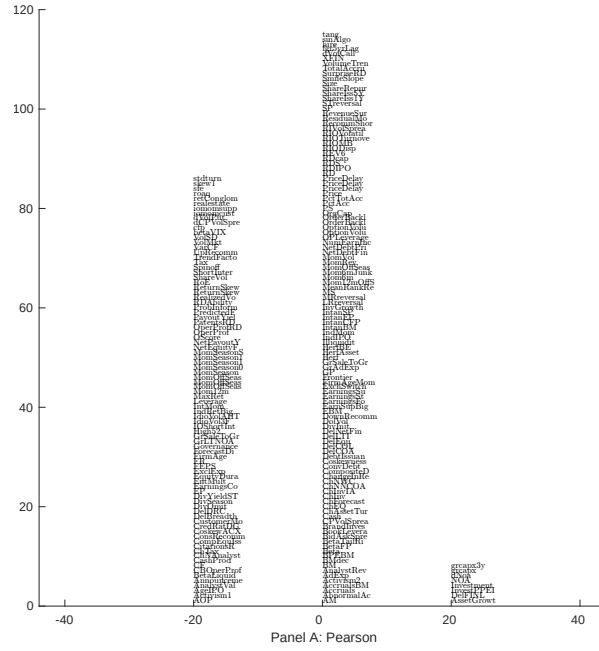


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CNS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

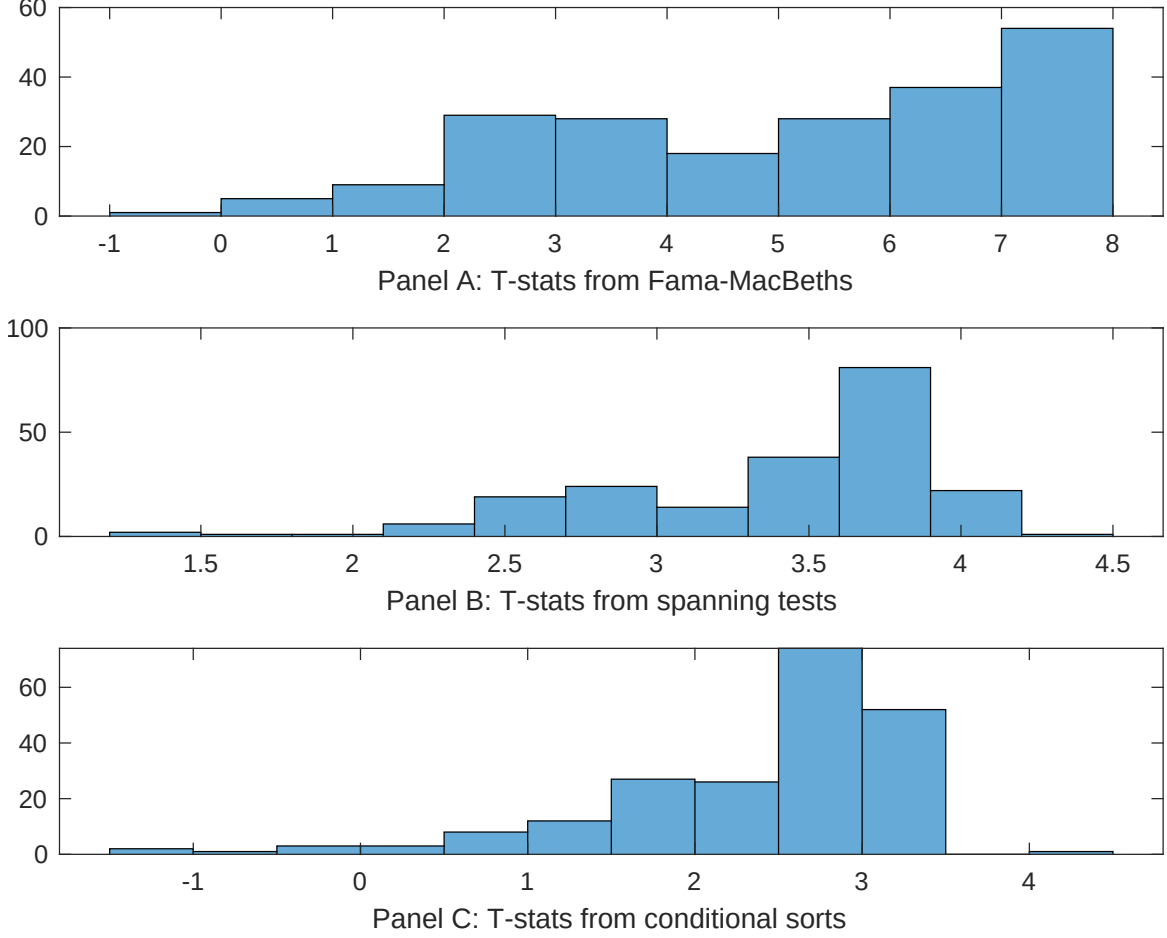


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CNS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.08]	0.13 [5.82]	0.12 [5.59]	0.17 [6.99]	0.14 [6.61]	0.13 [5.68]	0.15 [5.48]
CNS	0.63 [3.66]	0.70 [3.71]	0.11 [5.93]	0.94 [5.37]	0.87 [4.38]	0.90 [4.20]	0.20 [0.89]
Anomaly 1	0.16 [8.11]						0.20 [0.69]
Anomaly 2		0.13 [8.80]					1.00 [3.59]
Anomaly 3			0.89 [6.14]				-0.66 [-2.40]
Anomaly 4				0.80 [6.74]			0.52 [0.33]
Anomaly 5					0.14 [7.60]		0.83 [3.91]
Anomaly 6						0.52 [9.22]	0.67 [0.90]
# months	732	720	732	732	727	732	712
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CNS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CNS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24 [3.21]	0.21 [2.72]	0.19 [2.42]	0.19 [2.44]	0.26 [3.33]	0.23 [2.99]	0.20 [2.66]
Anomaly 1	28.61 [8.18]						23.26 [5.67]
Anomaly 2		14.10 [3.23]					1.46 [0.31]
Anomaly 3			19.83 [5.36]				10.62 [2.84]
Anomaly 4				17.65 [5.45]			10.85 [3.22]
Anomaly 5					6.61 [1.75]		-5.31 [-1.37]
Anomaly 6						8.42 [2.69]	-1.77 [-0.54]
mkt	-0.05 [-0.03]	0.38 [0.20]	0.64 [0.35]	-0.33 [-0.18]	0.47 [0.26]	0.19 [0.10]	0.17 [0.10]
smb	8.83 [3.42]	10.12 [3.78]	11.12 [4.19]	12.78 [4.75]	8.27 [3.00]	10.53 [3.91]	12.01 [4.42]
hml	-4.76 [-1.37]	-2.72 [-0.76]	-1.64 [-0.46]	-6.37 [-1.75]	-2.74 [-0.76]	-2.02 [-0.56]	-6.02 [-1.71]
rmw	-6.49 [-1.86]	-6.21 [-1.72]	-3.67 [-1.02]	-8.16 [-2.28]	-6.23 [-1.73]	-5.32 [-1.46]	-5.49 [-1.57]
cma	-8.98 [-1.57]	2.45 [0.40]	16.43 [3.14]	12.69 [2.44]	8.45 [1.51]	6.07 [1.03]	-0.93 [-0.14]
umd	0.11 [0.06]	-0.13 [-0.07]	-0.23 [-0.13]	-0.47 [-0.26]	-0.24 [-0.13]	-0.73 [-0.39]	-0.04 [-0.02]
# months	732	732	732	732	728	732	728
$\bar{R}^2(\%)$	12	5	7	7	4	5	13

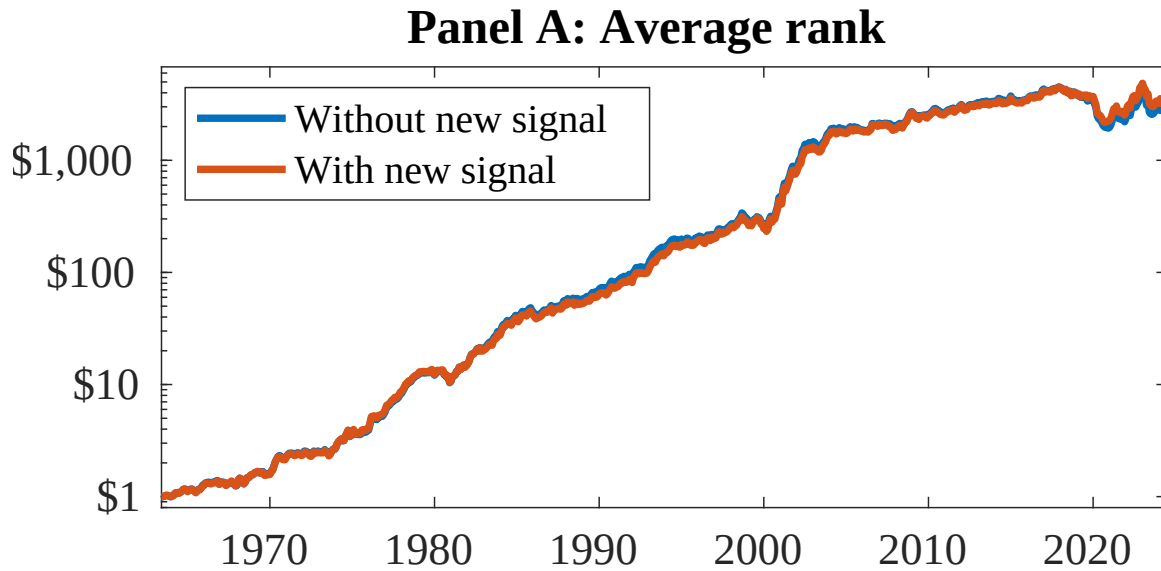


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CNS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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